EXERCISE 1 Without using a calculator, find :

1. $\csc \frac{\pi}{3}$
2. $\cot \frac{2\pi}{3}$
3. $\sec \frac{5\pi}{6}$
4. $\cot \pi$

EXERCISE 2 Without using a calculator, find $\csc x, \sec x,$ and $\cot x$ given :

1. $\sin x = \frac{3}{5}, \quad 0 \leq x \leq \frac{\pi}{2}$
2. $\cos x = \frac{2}{3}, \quad \frac{3\pi}{2} < x < 2\pi$

EXERCISE 3 Find the other *five trigonometric ratios* if :

1. $\cos x = \frac{3}{4}$ and $\frac{3\pi}{2} < x < 2\pi$
2. $\sin x = -\frac{2}{3}$ and $\pi < x < \frac{3\pi}{2}$
3. $\sec x = 2\frac{1}{2}$ and $0 < x < \frac{\pi}{2}$
4. $\csc x = 2$ and $\frac{\pi}{2} < x < \pi$
5. $\tan \beta = \frac{1}{2}$ and $\pi < \beta < \frac{3\pi}{2}$
6. $\cot \theta = \frac{4}{3}$ and $\pi < \theta < \frac{3\pi}{2}$

EXERCISE 4 Using the graph of $y = \cos x$, sketch the graph of

$$y = \frac{1}{\cos x} = \sec x$$

for $-2\pi \leq x \leq 2\pi$. Check your answer using technology.

EXERCISE 5 Using the graph of $y = \tan x$, sketch the graph of

$$y = \frac{1}{\tan x} = \cot x$$

for $-2\pi \leq x \leq 2\pi$.

EXERCISE 6 Use technology to sketch $y = \sec x$ and $y = \csc(x + \frac{\pi}{2})$ on the same set of axes for $-2\pi \leq x \leq 2\pi$. Explain your answer.

EXERCISE 7 Solve exactly for $0 \leq x \leq 2\pi$:

1. $\sec x = 2$
2. $\csc x = -\sqrt{2}$
3. $\sqrt{3} \sec 2x = -2$
4. $\csc(x + \frac{\pi}{6}) + \sqrt{2} = 0$
5. $\cot x + 1 = 0$
6. $\cot(2x - \frac{\pi}{4}) - \sqrt{3} = 0$

EXERCISE 8 Find, giving your answer in radians :

1. $\arccos 1$
2. $\arcsin(-1)$
3. $\arctan 1$
4. $\arctan(-1)$
5. $\arcsin \frac{1}{2}$
6. $\arccos\left(-\frac{\sqrt{3}}{2}\right)$
7. $\arctan \sqrt{3}$
8. $\arccos\left(-\frac{1}{\sqrt{2}}\right)$
9. $\arctan\left(-\frac{1}{\sqrt{3}}\right)$
10. $\sin^{-1}(-0.767)$
11. $\cos^{-1} 0.327$
12. $\tan^{-1}(-50)$

EXERCISE 9 Find the invariant point for the inverse transformation from :

1. $y = \sin x$ to $y = \arcsin x$
2. $y = \tan x$ to $y = \arctan x$
3. $y = \cos x$ to $y = \arccos x$

EXERCISE 10 Simplify :

1. $\arcsin(\sin \frac{\pi}{3})$
2. $\arccos(\cos(-\frac{\pi}{6}))$
3. $\tan(\arctan(0.3))$
4. $\cos(\arccos(-\frac{1}{2}))$
5. $\arctan(\tan \pi)$
6. $\arcsin(\sin \frac{4\pi}{3})$